Financial Database Management - BNP Paribas

AOUSTIN Manon, JIANG Elodie, LECLERC Rose, RACHIDI Zineb

Summary

01 Introduction & context

02 Regulatory framework

03 Technical Solution

04 Justification & Conclusion

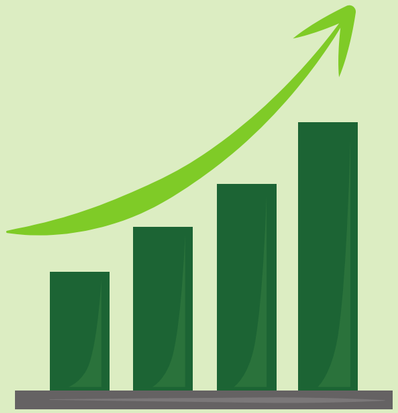
Issue

How can BNP Paribas build a real-time database to group risk data from 65 countries and follow BCBS 239 rules ?

Company Overview

BNP Paribas : A Global Financial Giant

Founded in 1848
65 Countries
185 000 Employees
€2.9 Trillion Assets (AUM 2024)



Status : G-SIB (Global Systemically Important Bank)
Tagline : "Too Big to Fail" = Strictest Regulations

Why Data Management is Critical ?

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The Data Challenge : From Silence to Compliance

Before 2008 crisis :
Scattered Information
Too Slow
Hidden Risks

After crisis :
BCBS 239 Rule : Accuracy & Speed.
Deadline : Reports in < 24 hours during a crisis

Target : One unified architecture for 65 countries

Regulatory context

Timeline

2008

Financial crisis

2013

BCBS 239 publication

2016

Entry in force for BNP Paribas

Regulatory context

The 4 pillars

P1-P2
Governance & Infrastructure

P3-P6
Data Aggregation

P7-P11
Risk Reporting

P12-P14
Regulatory oversight

DACT model

D — Data Accuracy → error < 0.01%

A — Adaptability → reports < 24h

C — Completeness → coverage ≥ 99.5%

T — Timeliness → data available T+0 to T+1

What data BNP Paribas need to manage ?

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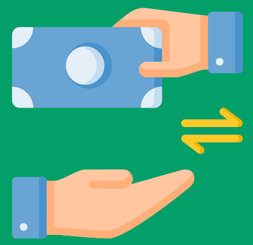
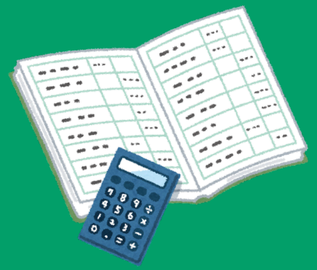
Data Category	Volume	Frequency	Example
Market Data (prices)	Millions of records/day	Real-time	equity prices, bond yields
Trading Positions	Hundreds of thousands	Intraday	Long/Short equity, derivatives
Credit Exposure	Tens of thousands	Daily (T+1)	Counterparty risk
Regulatory Reports	Dozens of report types	Daily / Weekly	VaR, stress tests, ICAAP reports





Technical Solution

Type	Exemple	Strengths	Weaknesses
Relational (RDBMS)	PostgreSQL	Data Integrity (ACID), structured, regulatory-grade, audit trails	Less scalable for unstructured data
Columnar / Analytical	Snowflake	Fast aggregations on large datasets, cloud-native	Not optimized for transactional writes
In-Memory	Redis, Apache Ignite	Ultra-low Latency, real-time	Volatile, expensive at scale
Document / NoSQL	MongoDB	Flexible schema, handles unstructured/alternative data	Weaker consistency guarantees

Our recommendation:

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Layer 1

Transactional — golden source PostgreSQL
Stores: positions, counterparty data, legal entity hierarchy, reference data
Why: ACID guarantees, full audit trails

Layer 2

Analytical — aggregation engine Snowflake
Stores: historical risk metrics, P&L data, stress test results, regulatory reports
Why: Columnar storage for fast aggregations

Layer 3

Real-time — in-memory cache Redis
Stores: live market prices, intraday VaR, real-time exposure snapshots
Why: Sub-millisecond latency, pub/sub for live feeds

Data flows with Apache Kafka

Architecture Aligned with BCBS 239

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ACCURACY

PostgreSQL

- ACID compliance

TIMELINESS

Kafka & Redis

- Real time streaming
- Instant monitoring

ADAPTABILITY & COMPLETENESS

Snowflake

- Fast aggregation
- Across 65 countries

From Data to Decision

Master Data

Analytical and Risky Data

Position ID	Asset Class	Counterparty	Country	Notional EUR	Market Value	Rating	VaR Contribution
BNPP_TX_998	Corporate Bond	L'Oréal	FR	5,000,000	6,900,000	AA	12,000
BNPP_TX_999	Sovereign	French State	US	12,000,000	12,000,500	AAA	2,1000
BNPP_TX_001	Derivative	JP Morgan	US	2,500,000	-150,000	A+	45,000
BNPP_TX_003	Equity	HSBC	GB	3,800,000	3,900,000	A+	16,800
BNPP_TX_003	FX	Deutsche Bank	DE	8,800,000	8,000,000	AA-	9,800
BNPP_TX_004	Credit Default Swap	Goldman Sachs	US	1,200,000	-280,000	A+	67,000

Market Value

Conclusion

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Thank you for your attention !